Technical Report: Disk Imaging, Digital Forensic & Memory Artifacts Analysis

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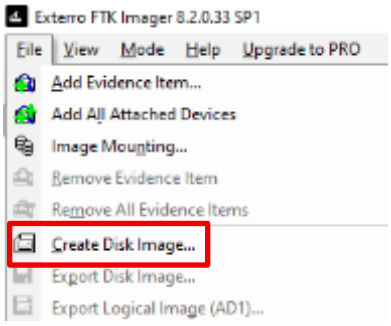
I. Part 1: Data Acquisition

Q1. Static Data Acquisition from a USB Disk

a. Using FTK Imager data acquisition tool

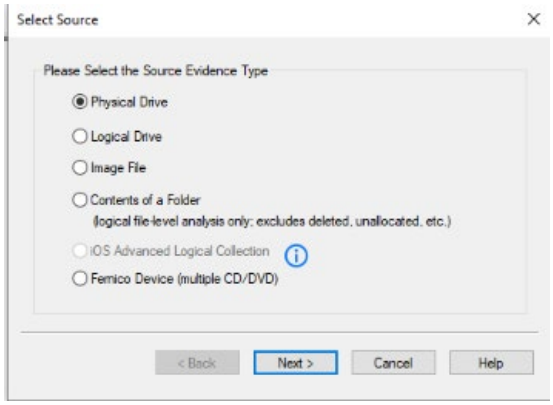
Step 1: Start Image creation.

Click on File, then select Create Disk Image Source Data Selection.

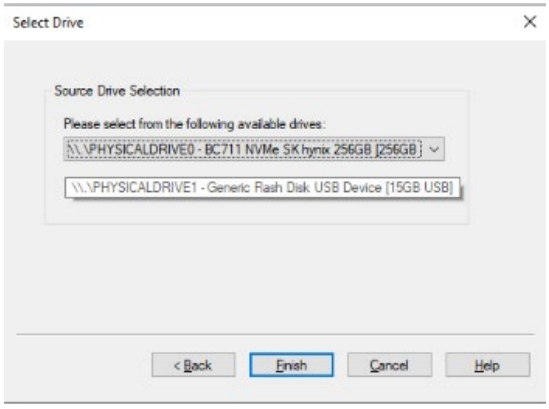


Step 2: Select Source

For source type, select Physical Drive for examination of full hard drive.

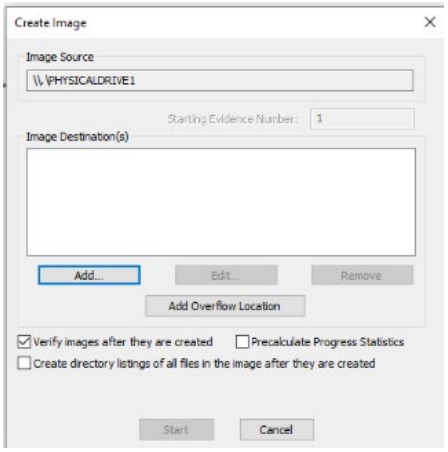


In the dropdown list, select the USB drive.

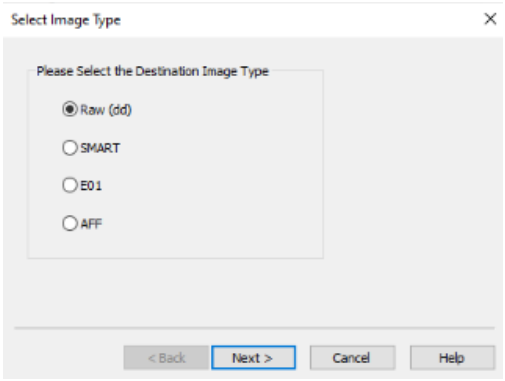


Step 3: Select Image Type and Destination

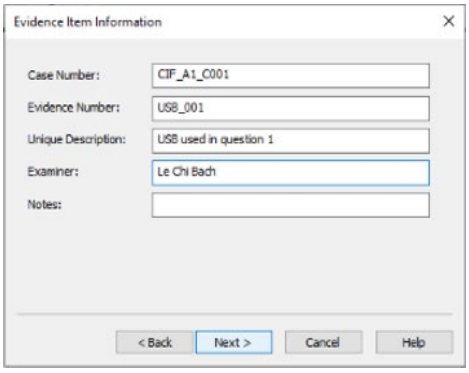
In Create Image Window, select Add to configure image type and destination.



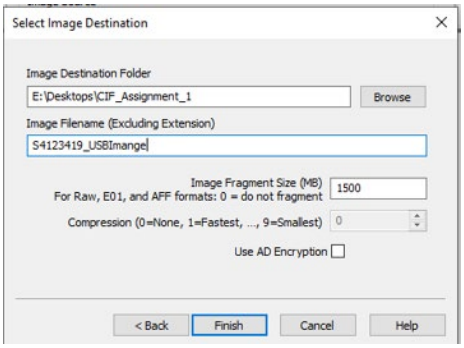
Select image type (Raw(dd) as required)



Enter evidence information

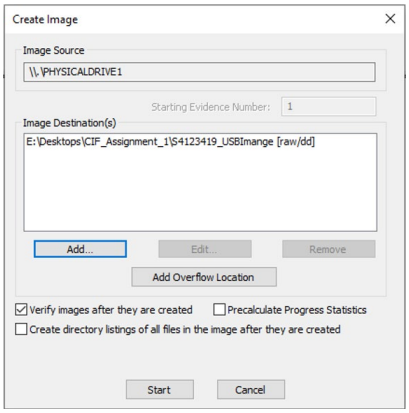


Set image destination, name the file as required

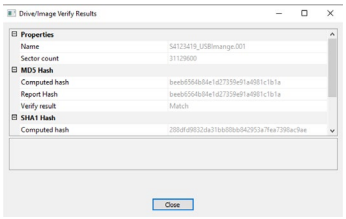


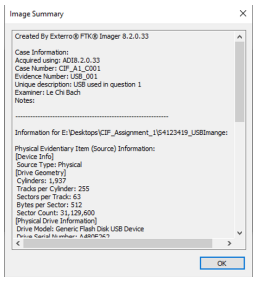
Step 4: Start the process and verify

Make sure the Verify images after they are created box is ticked and click Start



After the process is done, see the result and summary





b. Provide the details of the disk image information generated by FTK Imager

Created By Exterro® FTK® Imager 8.2.0.33

Case Information:
Acquired using: ADI8.2.0.33
Case Number: CIF_A1_C001
Evidence Number: USB_001
Unique description: USB used in question 1
Examiner: Le Chi Bach
Notes:

Information for E:\Desktops\CIF_Assignment_1\S4123419_USBImange:

Physical Evidentiary Item (Source) Information:
[Device Info]
Source Type: Physical
[Drive Geometry]
Cylinders: 1,937
Tracks per Cylinder: 255
Sectors per Track: 63
Bytes per Sector: 512
Sector Count: 31,129,600
[Physical Drive Information]
Drive Model: Generic Flash Disk USB Device
Drive Serial Number: A480F262
Drive Interface Type: USB
Removable drive: True
Source data size: 15200 MB
Sector count: 31129600
[Computed Hashes]
MD5 checksum: beeb6564b84e1d27359e91a4981c1b1a
SHA1 checksum: 288dfd9832da31bb88bb842953a7fea7398ac9ae

Image Information:
Acquisition started: Fri Mar 13 14:57:14 2026
Acquisition finished: Fri Mar 13 15:17:28 2026
Segment list:

E:\Desktops\CIF_Assignment_1\S4123419_USBImange.001

E:\Desktops\CIF_Assignment_1\S4123419_USBImange.002

E:\Desktops\CIF_Assignment_1\S4123419_USBImange.003

E:\Desktops\CIF_Assignment_1\S4123419_USBImange.004

E:\Desktops\CIF_Assignment_1\S4123419_USBImange.005

E:\Desktops\CIF_Assignment_1\S4123419_USBImange.006

E:\Desktops\CIF_Assignment_1\S4123419_USBImange.007

E:\Desktops\CIF_Assignment_1\S4123419_USBImange.008

E:\Desktops\CIF_Assignment_1\S4123419_USBImange.009

E:\Desktops\CIF_Assignment_1\S4123419_USBImange.010

E:\Desktops\CIF_Assignment_1\S4123419_USBImange.011
COMPUTED HASH : beeb6564b84e1d27359e91a4981c1b1a
COMPUTED HASH : 288dfd9832da31bb88bb842953a7fea7398ac9ae

Image Verification Results:
Verification started: Fri Mar 13 15:17:28 2026
Verification finished: Fri Mar 13 15:18:07 2026
MD5 checksum: beeb6564b84e1d27359e91a4981c1b1a : verified
SHA1 checksum: 288dfd9832da31bb88bb842953a7fea7398ac9ae : verified

c. Answering questions

When was the disk image created?

- The disk image was created on Friday March 13th, 2026, at 14:57:14.

Show at least one checksum information of the physical disk.

- MD5 checksum: beeb6564b84e1d27359e91a4981c1b1a

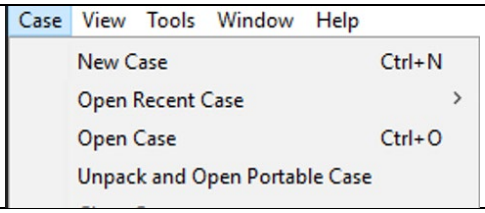
What was the checksum of the image? Was the verification of checksum successful?

- MD5 checksum: beeb6564b84e1d27359e91a4981c1b1a. Verified successful.
- SHA1 checksum: 288dfd9832da31bb88bb842953a7fea7398ac9ae. Verified successful.

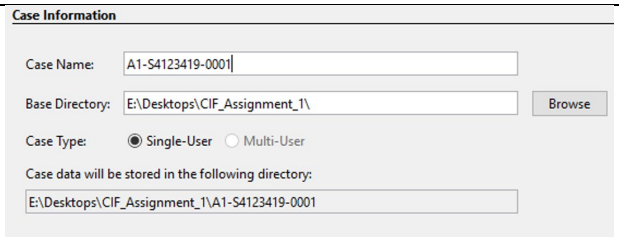
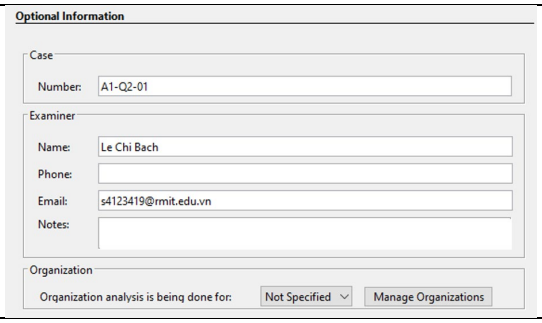
Q2. Analyzing Disk Image using Autopsy Tool

a. Creating a case using Autopsy Tool

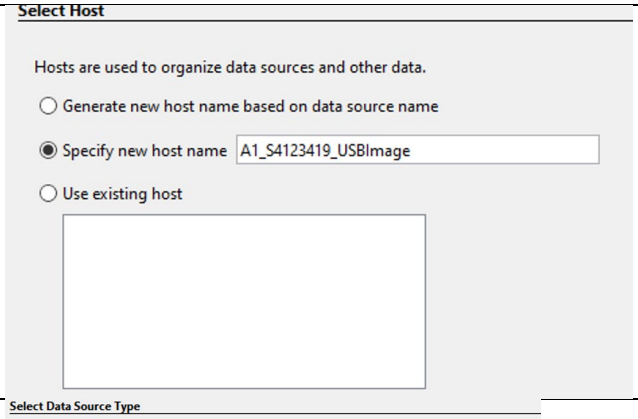
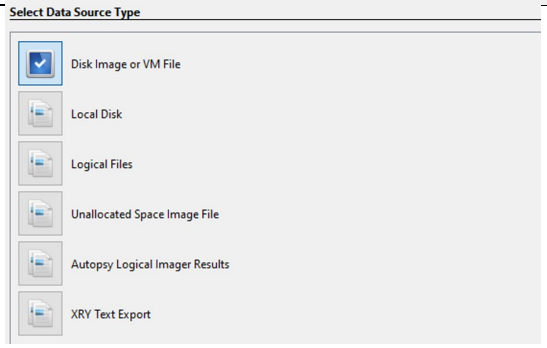
Step 1: Start case creation.

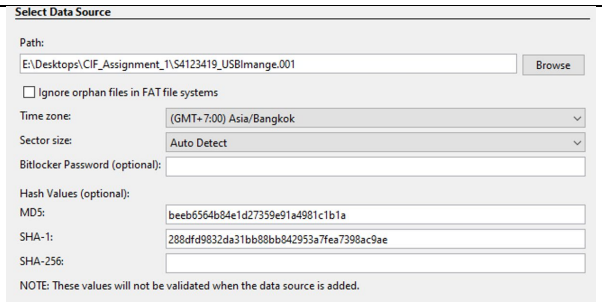
Click on Case, then New Case.	

Step 2: Set case information

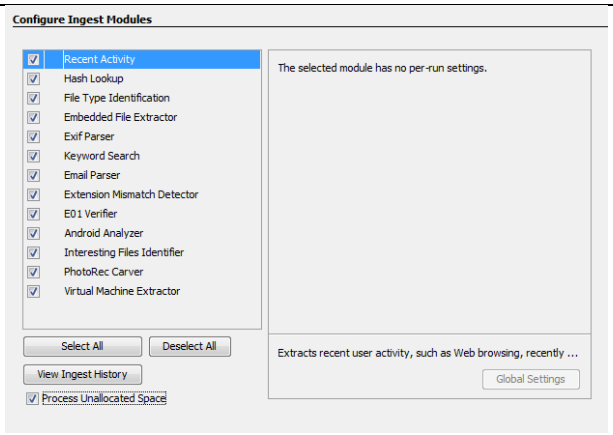
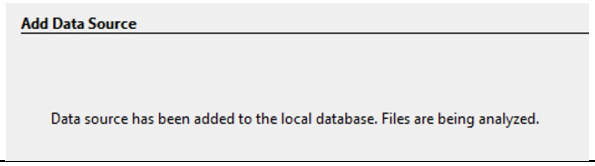
Set case name, base directory, choose single-user case type for this activity is done individually.	
For optional information, give the case a name, fill in examiner information and specify the organization for whom this case is done for.	

Step 3: Add data source:

Select a type of host (Specify new host name as the requirements)	
For data source type, select Disk Image or VM File because a RAW/dd file is being examined.	

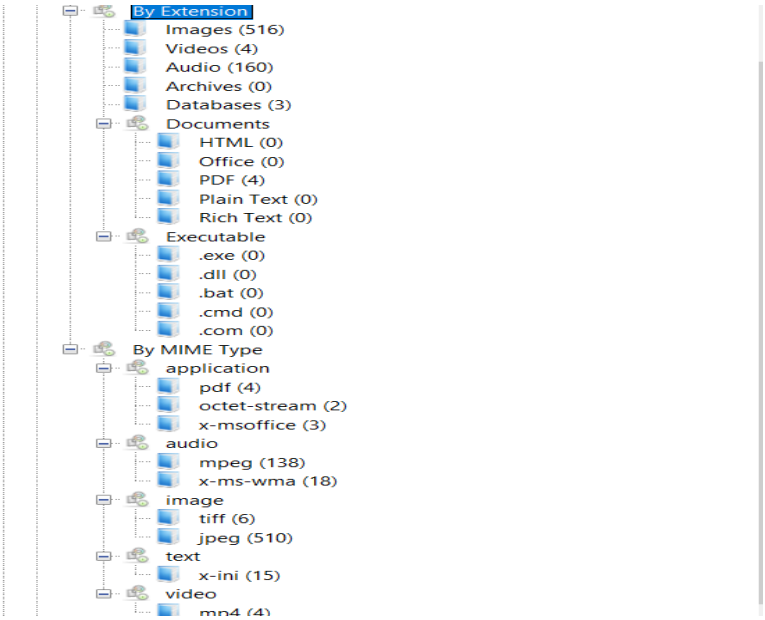
For data source, click on Browse and choose the evidence file (the disk image created in Question 1). Provides the hash value found in Question 1 for validating evidence integrity.	

Step 4: Configure inspect module

Select all the necessary modules to analyze the evidence. Click Next, then Finish then wait till the process is done.	 

b. Show the list of different types of files and their numbers found in the disk image.

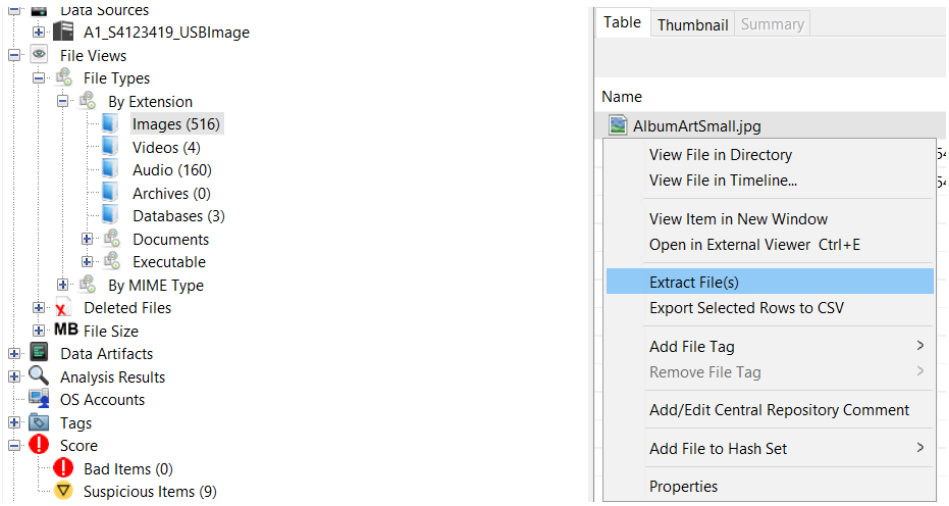
File type (by Extension)	Number	File type (by MIME Type)	Number
Images	516	application	pdf(4), octet-stream(2), x-msoffice(3)
Videos	4	audio	mpeg(138), x-ms-wma(18)
Audio	160	image	tiff(6), jpeg (510)
Archives	0	text	x-ini(15)
Database	3	video	mp4(4)
Documents	4		
Executable	0		



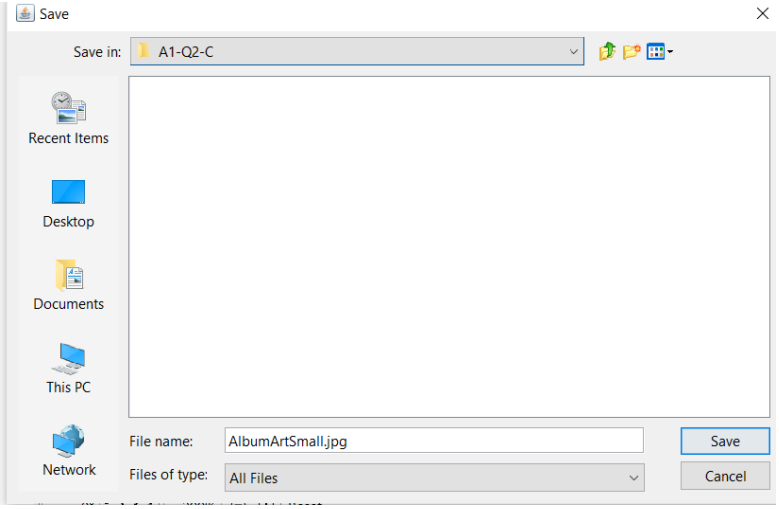
This is the file tree showing all the files, types of files and their number.

c. Demonstrate the extraction process of a file with a step-by-step description with screenshots

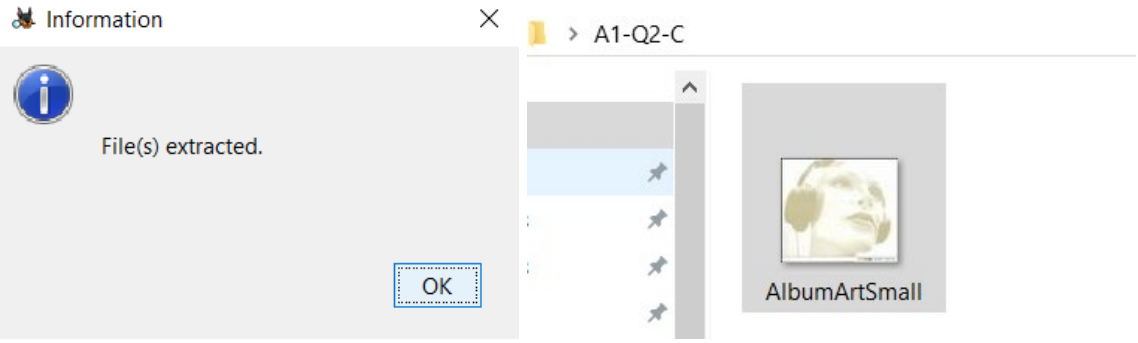
Step 1: Choose a file to extract and right click on it. Select Extract Files(s).



Step 2: Choose a destination and click Save



Step 3: Wait for the confirmed message and check for the file in the designated destination.



Q3. Understanding of Different Digital Forensics Tools

a. Static data acquisition tools:

- EnCase Forensic (Windows): EnCase Forensic is a professional digital forensics tool used mainly by law enforcement and corporate investigators. It supports data acquisition, analysis, and reporting, and is well known for its strong legal credibility and ability to maintain evidence integrity using hashing [1].
    - Advantages:
        - Widely accepted in courts [2]
        - Support remote acquisition [2]
    - Disadvantages:
        - Expensive license cost [1]
        - Steep learning curve for beginners [2]
- Guymager (Linux): Guymager is a forensic acquisition program capable of creating disk images in different formats [3].
    - Advantages:
        - Free and open source [4]
        - Easy user interface [4]
    - Disadvantages:

        - metadata limitations [5]
        - error handling and configuration complexity [5]
- Cellebrite Digital Collector (MacOS): Cellebrite Digital Collector is a tool for creating physical decrypted image for Apple's Mac and Windows computer.
    - Advantages:
        - Easy create forensic image [6]
        - Targeted data collection. [6]
    - Disadvantages:
        - High-cost commercial tool.

b. Memory acquisition tools:

- DumpIt (Windows): DumpIt is a fast memory acquisition tool for Windows, generates full memory crash dumps of Windows machines [7].
    - Advantages:
        - Easy to deploy [7]
        - Fast memory acquisition [7]
    - Disadvantages:
        - Incompatible with older 64-bit Windows versions [8]
- LiME (Linux): LiME (Linux Memory Extractor) allows the acquisition of volatile memory from Linux and Linux-based devices [9]
    - Advantages:
        - Acquisition over network interface or to local disk [9]
        - Allows multiple output format [9]
    - Disadvantages:
        - Requires compile kernel, make it inconvenient to use. [10]
- OSX Memory Imager: OSX Memory Imager is an open-source tool to acquire physical memory on an Intel based Mac. [11]
    - Advantages:
        - Open-source availability [11]
    - Disadvantages:
        - Does not allow access to memory ports or memory-mapped I/O devices [12]

II. Part 2: Forensic Analysis

Q4. Disk Image Artifact Analysis

a. File Listing and carving

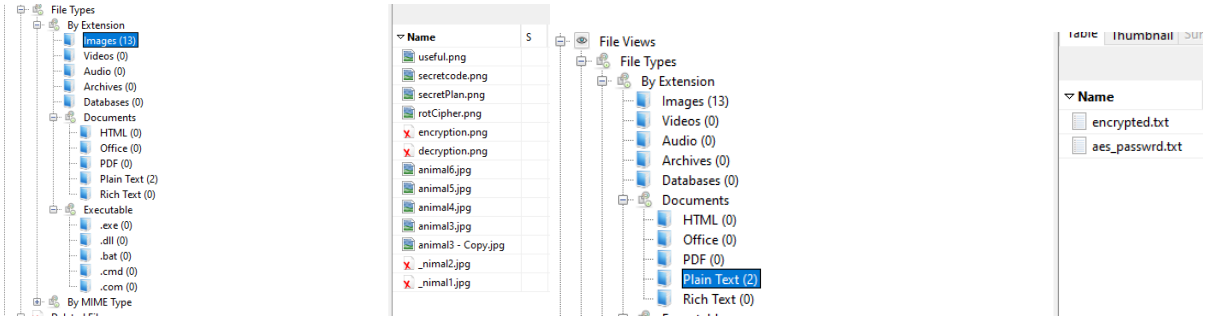
After analyzing the disk image, the required tasks can be done.

I. All the files presented on the drive:

useful.png	secretcode.png	animal6.jpg
secretPlan.png	rotCipher.png	animal5.jpg

animal4.jpg animal3 - Copy.jpg aes_password.txt.

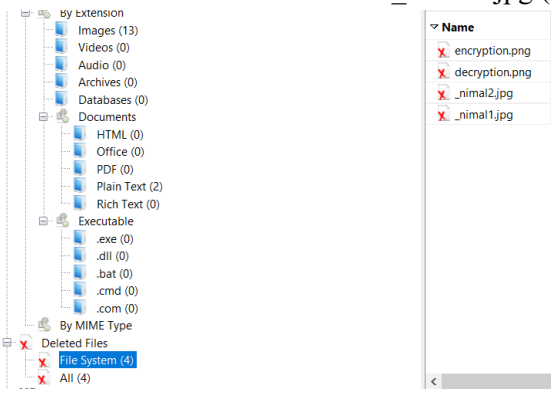
animal3.jpg encrypted.txt



II. Identify and list all files that were deleted before recovery, including their file types

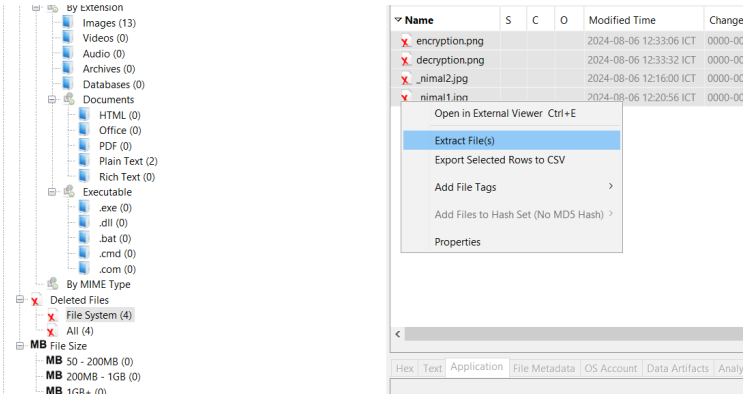
encryption.png (Image) _nimal2.jpg (Image)

decryption.png (Image) _nimal1.jpg (Image)

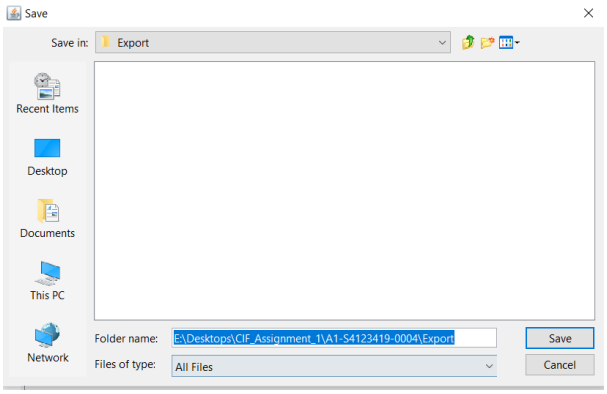


III. Recover all image files from the disk image

In the Deleted Files folder, all the image files are selected, right click and select Extract File(s)

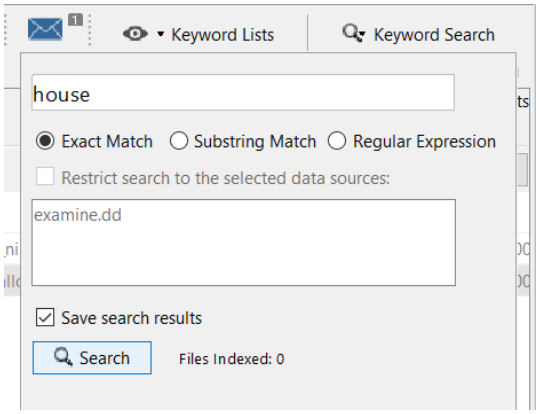


Choose a destination and click Save.

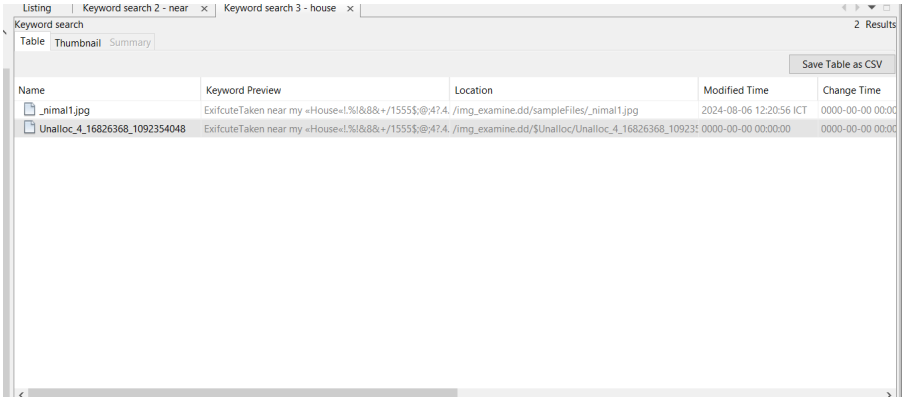


b. Finding Suspect's hideout location

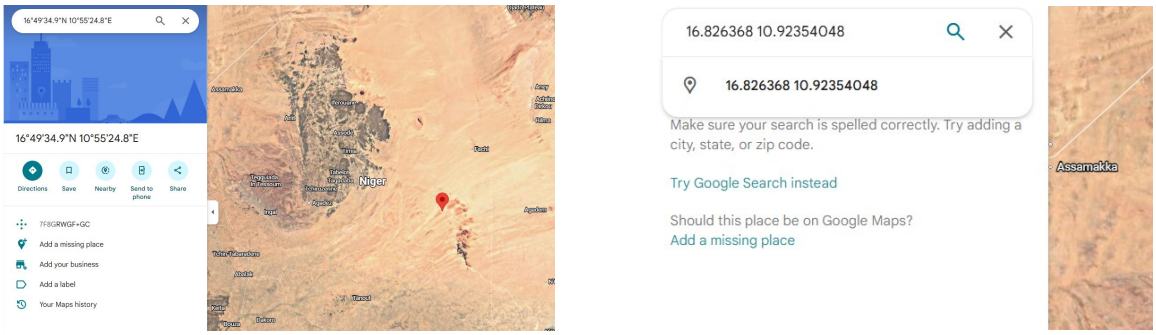
Use the Keyword Search functionality to search for "near" or "house".



There are 2 results with one file named "Unalloc_4_16826368_1092354048".



Copy "16826368_1092354048" change it to "16.826368 10.92354048" and paste it to Google Maps, the suspect's hideout can be identified. It is somewhere in Niger.



Maps showing suspect's hideout

Q5. Memory Data Analysis from Computer Memory

a. Show memory dump basic information

- Command: python3 vol.py -f /home/kali/Desktop/xp-laptop-2005-07-04-1430.img windows.info
- Result:

```
Kernel Base     0×804d7000
DTB     0×39000
Symbols file:///root/volatility3/volatility3/symbols/windows/ntoskrnl.pdb/329
62337F0F646388B39535CD8DD70E8-2.json.xz
Is64Bit False
IsPAE   False
layer_name      0 WindowsIntel
memory_layer    1 FileLayer
KdDebuggerDataBlock     0×8054c060
NTBuildLab      2600.xpsp_sp2_gdr.050301-1519
CSDVersion      2
KdVersionBlock  0×8054c038
Major/Minor     15.2600
MachineType     332
KeNumberProcessors      1
SystemTime      2005-07-04 18:30:32+00:00
NtSystemRoot    C:\WINDOWS
NtProductType   NtProductWinNt
NtMajorVersion  5
NtMinorVersion  1
PE MajorOperatingSystemVersion  5
PE MinorOperatingSystemVersion  1
PE Machine      332
PE TimeDateStamp        Wed Mar  2 00:59:37 2005
```

Memory dump basic information

b. Show the list of all running processes.

- Command: python3 vol.py -f /home/kali/Desktop/xp-laptop-2005-07-04-1430.img windows.pslist
- Result:

```
PID     PPID    ImageFileName   Offset(V)       Threads Handles SessionId   W
ow64    CreateTime      ExitTime        File output

4       0       System  0×823c87c0      62      1133    N/A     False   N/A N
/A      Disabled
400     4       smss.exe        0×8214b020      3       21      N/A     False
2005-07-04 18:17:26.000000 UTC  N/A     Disabled
456     400     csrss.exe       0×821c11a8      11      551     0       False
2005-07-04 18:17:29.000000 UTC  N/A     Disabled
480     400     winlogon.exe    0×814dc020      18      522     0       False
2005-07-04 18:17:29.000000 UTC  N/A     Disabled
524     480     services.exe    0×815221c8      17      321     0       False
2005-07-04 18:17:30.000000 UTC  N/A     Disabled
536     480     lsass.exe       0×821d8248      20      369     0       False
2005-07-04 18:17:30.000000 UTC  N/A     Disabled
680     524     svchost.exe     0×814f0020      19      206     0       False
2005-07-04 18:17:31.000000 UTC  N/A     Disabled
760     524     svchost.exe     0×821daa88      10      289     0       False
2005-07-04 18:17:31.000000 UTC  N/A     Disabled
800     524     svchost.exe     0×821463a8      75      1558    0       False
2005-07-04 18:17:31.000000 UTC  N/A     Disabled
840     524     Smc.exe 0×8216c9b0      22      421     0       False   2005-
07-04 18:17:32.000000 UTC       N/A     Disabled
932     524     svchost.exe     0×81530228      6       93      0       False
2005-07-04 18:17:33.000000 UTC  N/A     Disabled
972     524     svchost.exe     0×81534c10      15      212     0       False
2005-07-04 18:17:34.000000 UTC  N/A     Disabled
1104    524     spoolsv.exe     0×8202e7e8      11      145     0       False
2005-07-04 18:17:38.000000 UTC  N/A     Disabled
1272    524     ati2evxx.exe    0×8152f9a0      4       38      0       False
2005-07-04 18:17:39.000000 UTC  N/A     Disabled
1356    524     Crypserv.exe    0×820ac020      3       34      0       False
2005-07-04 18:17:40.000000 UTC  N/A     Disabled
1380    524     DefWatch.exe    0×81521da0      3       27      0       False
2005-07-04 18:17:40.000000 UTC  N/A     Disabled
1440    524     msdtc.exe       0×820b5670      15      164     0       False
```

Running processes

c. Show the list of some of the processes with detailed information.

- Command:python3 vol.py -f /home/kali/Desktop/xp-laptop-2005-07-04-1430.img windows.psscan
- Result:

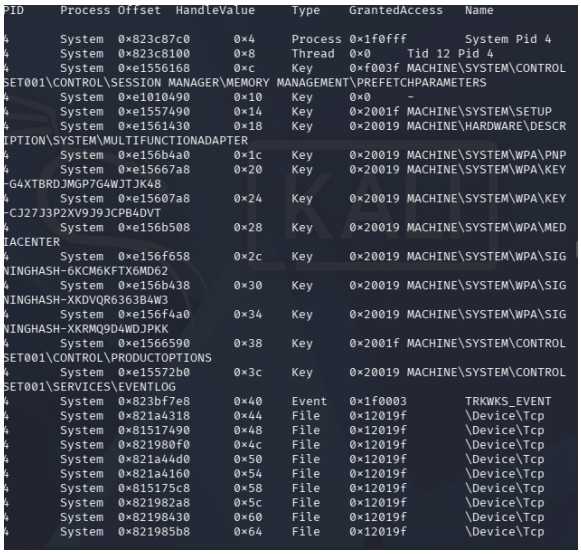
```
Volatility 3 Framework 2.28.0
Progress:  100.00               PDB scanning finished
PID     PPID    ImageFileName   Offset(V)       Threads Handles SessionId   W
ow64    CreateTime      ExitTime        File output

3612    3352    PluckTray.exe   0×13383b0       3       102     0       False2
005-07-04 18:24:00.000000 UTC   N/A     Disabled
3276    2392    firefox.exe     0×133d810       7       189     0       False2
005-07-04 18:21:11.000000 UTC   N/A     Disabled
2496    2392    VPTray.exe      0×1462be0       2       111     0       False2
005-07-04 18:18:06.000000 UTC   N/A     Disabled
2392    2300    explorer.exe    0×146e860       18      489     0       False2
005-07-04 18:18:03.000000 UTC   N/A     Disabled
3128    800     wuauclt.exe     0×1474510       3       157     0       False2
005-07-04 18:19:11.000000 UTC   N/A     Disabled
368     3352    PluckUpdater.ex 0×1488350       0       -       0       False2
005-07-04 18:24:30.000000 UTC   2005-07-04 18:26:44.000000 UTC  Disabled
2692    2392    WZQKPICK.EXE    0×14b8a58       1       17      0       False2
005-07-04 18:18:15.000000 UTC   N/A     Disabled
480     400     winlogon.exe    0×14dc020       18      522     0       False2
005-07-04 18:17:29.000000 UTC   N/A     Disabled
2548    2392    jusched.exe     0×14ecc00       1       22      0       False2
005-07-04 18:18:07.000000 UTC   N/A     Disabled
680     524     svchost.exe     0×14f0020       19      206     0       False2
005-07-04 18:17:31.000000 UTC   N/A     Disabled
1380    524     DefWatch.exe    0×1521da0       3       27      0       False2
005-07-04 18:17:40.000000 UTC   N/A     Disabled
524     480     services.exe    0×15221c8       17      321     0       False2
005-07-04 18:17:30.000000 UTC   N/A     Disabled
1272    524     ati2evxx.exe    0×152f9a0       4       38      0       False2
005-07-04 18:17:39.000000 UTC   N/A     Disabled
932     524     svchost.exe     0×1530228       6       93      0       False2
005-07-04 18:17:33.000000 UTC   N/A     Disabled
972     524     svchost.exe     0×1534c10       15      212     0       False2
005-07-04 18:17:34.000000 UTC   N/A     Disabled
3256    2392    cmd.exe 0×153f480       1       29      0       False   2005-0
7-04 18:20:58.000000 UTC        N/A     Disabled
2472    2392    TaskSwitch.exe  0×1540da0       1       24      0       False2
005-07-04 18:18:05.000000 UTC   N/A     Disabled
3300    3256    dd.exe  0×1543870       1       22      0       False   2005-0
7-04 18:30:32.000000 UTC        N/A     Disabled
```

Processses with detailed information

d. Show the list of handles of processes

- Command:python3 vol.py -f /home/kali/Desktop/xp-laptop-2005-07-04-1430.img windows.handles
- Result:



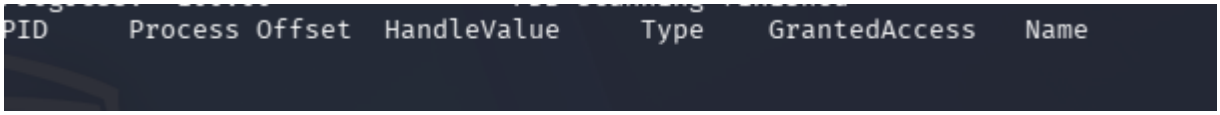
```
PID     Process Offset  HandleValue     Type    GrantedAccess   Name

4       System  0x823c87c0      0x4     Process 0x1f0fff        System Pid 4
4       System  0x823c8100      0x8     Thread  0x0     Tid 12 Pid 4
4       System  0xe1556168      0xc     Key     0xf003f MACHINE\SYSTEM\CONTROL
SET001\CONTROL\SESSION MANAGER\MEMORY MANAGEMENT\PREFETCHPARAMETERS
4       System  0xe1010490      0x10    Key     0x0     -
4       System  0xe1557490      0x14    Key     0x2001f MACHINE\SYSTEM\SETUP
4       System  0xe1561430      0x18    Key     0x20019 MACHINE\HARDWARE\DESCR
IPTION\SYSTEM\MULTIFUNCTIONADAPTER
4       System  0xe156b4a0      0x1c    Key     0x20019 MACHINE\SYSTEM\WPA\PNP
4       System  0xe15667a8      0x20    Key     0x20019 MACHINE\SYSTEM\WPA\KEY
-G4XTBRDJMGP7G4WJTJK48
4       System  0xe15607a8      0x24    Key     0x20019 MACHINE\SYSTEM\WPA\KEY
-CJ27J3P2XV9J9JCPB4DVT
4       System  0xe156b508      0x28    Key     0x20019 MACHINE\SYSTEM\WPA\MED
IACENTER
4       System  0xe156f658      0x2c    Key     0x20019 MACHINE\SYSTEM\WPA\SIG
NINGHASH-6KCM6KFTX6MD62
4       System  0xe156b438      0x30    Key     0x20019 MACHINE\SYSTEM\WPA\SIG
NINGHASH-XKDVQR6363B4W3
4       System  0xe156f4a0      0x34    Key     0x20019 MACHINE\SYSTEM\WPA\SIG
NINGHASH-XKRMQ9D4WDJPKK
4       System  0xe1566590      0x38    Key     0x2001f MACHINE\SYSTEM\CONTROL
SET001\CONTROL\PRODUCTOPTIONS
4       System  0xe15572b0      0x3c    Key     0x20019 MACHINE\SYSTEM\CONTROL
SET001\SERVICES\EVENTLOG
4       System  0x823bf7e8      0x40    Event   0x1f0003        TRKWKS_EVENT
4       System  0x821a4318      0x44    File    0x12019f        \Device\Tcp
4       System  0x81517490      0x48    File    0x12019f        \Device\Tcp
4       System  0x821980f0      0x4c    File    0x12019f        \Device\Tcp
4       System  0x821a44d0      0x50    File    0x12019f        \Device\Tcp
4       System  0x821a4160      0x54    File    0x12019f        \Device\Tcp
4       System  0x815175c8      0x58    File    0x12019f        \Device\Tcp
4       System  0x821982a8      0x5c    File    0x12019f        \Device\Tcp
4       System  0x82198430      0x60    File    0x12019f        \Device\Tcp
4       System  0x821985b8      0x64    File    0x12019f        \Device\Tcp
```

List of handles of processes

e. Show the handles of the process with PID=1056.

- Command: python3 vol.py -f /home/kali/Desktop/xp-laptop-2005-07-04-1430.img windows.handles --pid 1056
- Result:

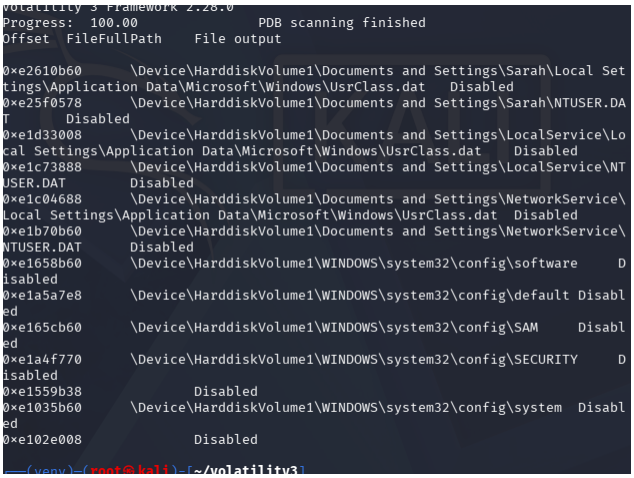


```
PID     Process Offset  HandleValue     Type    GrantedAccess   Name
```

Handles of process with PID = 1056 (no open handle)

f. Show the registry hives

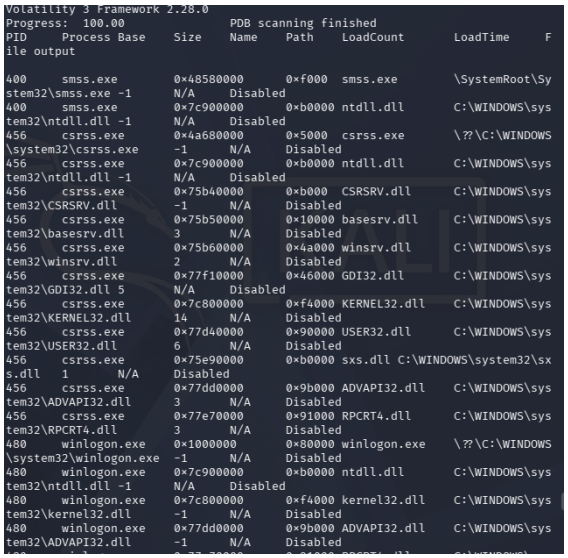
- Command: python3 vol.py -f /home/kali/Desktop/xp-laptop-2005-07-04-1430.img windows.registry.hivelist
- Result:



Registry hives

g. Show the list of Dynamic Link Libraries (DLLs) loaded by the processes.

- Command: python3 vol.py -f /home/kali/Desktop/xp-laptop-2005-07-04-1430.img windows.dlllist
- Result:



List of DLLs loadedby the processes

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